

Integrated Dual Degree
in
Computer Science & Engineering and Artificial Intelligence

NewGenHealthAI

An AI-Powered Medical Consultation System

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Abstract

NewGenHealthAI is a multi-modal AI-powered medical consultation system developed as a final project for the Integrated Dual Degree programme at RGIPT. The system brings together several modern AI technologies into one platform — including deep learning-based image classification, a large language model (LLM) for conversations, retrieval-augmented generation (RAG) for evidence-based answers, and a multi-agent workflow engine.

The project introduces three key ideas that make it different from regular medical chatbots:

- **Clinical Triage Agent** — Instead of giving an answer right away, the system asks the patient a series of follow-up questions, just like a real doctor would. It collects details like duration, severity, and associated symptoms before forming a diagnosis.
- **CLIP-Based Domain Routing** — Before analysing a medical image, the system first identifies what type of image it is (skin, X-ray, retina, etc.) using a powerful vision-language model called CLIP. It then sends the image only to the correct specialist model, ensuring accurate results.
- **Multi-Model Image Classification** — Four specialist deep learning models are used for disease classification, together covering 342 disease classes across skin, X-ray, retinal, and general medical images.

The backend uses a 70-billion-parameter language model (LLaMA 3.3 70B) served via Groq, a ChromaDB vector database for medical knowledge retrieval, and a LangGraph state machine to coordinate nine specialised AI agents. The frontend is a clean, responsive React application with a dark medical theme.

This system is purely a research and educational tool. It is not a certified medical device and should not be used to replace professional healthcare advice.

1. Introduction

Artificial Intelligence has made remarkable progress in healthcare over the past decade. Systems can now detect eye diseases from retinal scans, identify skin cancer from photographs, and answer basic medical questions through chatbots. However, most of these systems work in isolation. They either classify images or answer questions — very few do both together in a meaningful way.

Another common problem with AI medical chatbots is that they give instant answers to vague descriptions. If a user types "I have a headache", a typical chatbot might reply with a general paragraph about headaches. But that is not how a real doctor works. A doctor first asks: How long have you had it? How severe is it? Any other symptoms? Only after collecting this information does the doctor suggest a possible diagnosis.

NewGenHealthAI was built to address both of these problems. It is a unified system that can handle text-based symptom discussions, perform medical image analysis, retrieve relevant information from a medical knowledge base, and conduct a structured clinical interview — all within one application.

The project was developed under the guidance of Dr. Susham Biswas, Associate Professor and Former-Head of Department (CSE), RGIPT, and named in his honour to acknowledge his mentorship and contribution to the department's AI curriculum.

1.1 Motivation

Access to timely medical consultation is a challenge for millions of people, particularly in rural areas. While AI cannot replace a doctor, it can serve as a first point of contact — helping users understand their symptoms, guiding them on whether to seek urgent care, and providing general health information in an easy-to-understand format.

Existing medical AI tools typically fall into one of two categories: image classifiers (which analyse scans but do not converse) or chatbots (which talk but cannot look at images). This project bridges that gap by creating a system that does both — and does so in a thoughtful, clinically-inspired way.

2. Problem Statement

Current medical AI systems face three major limitations that this project aims to address:

2.1 One-Shot Responses

Most AI health chatbots respond to a symptom description with a single, generic answer. They do not follow up with clarifying questions. This is unreliable because the same symptom — a headache, for example — can point to dozens of different conditions depending on factors like duration, severity, and accompanying symptoms. Without gathering this information, any diagnosis is essentially a guess.

2.2 Isolated Modalities

Image classification models and conversational AI systems are usually developed and used separately. A user who wants both text-based advice and image analysis has to use different tools. There is no unified platform that intelligently handles both types of input in one conversation.

2.3 Cross-Domain Classification Errors

When multiple specialist image models are available, there is a risk of sending the wrong image to the wrong model. For example, a skin photograph processed by an X-ray model might incorrectly be labelled as a fracture. Without a domain-checking step, this type of error is difficult to prevent.

NewGenHealthAI addresses all three of these problems through its Clinical Triage Agent, unified multi-modal pipeline, and CLIP-based domain routing system.

3. Objectives

The main objectives of this project are:

1. To build a conversational AI system that asks follow-up questions before giving a diagnosis, replicating the clinical interview process used by real physicians.
2. To implement a multi-domain medical image classification pipeline that correctly identifies the type of image before running any specialist model.
3. To integrate retrieval-augmented generation (RAG) so that AI responses are grounded in real medical documents, reducing the risk of hallucinated information.
4. To develop a clean, responsive frontend interface that makes the system easy to use for non-technical users.
5. To demonstrate how multiple AI techniques — LLMs, CNNs, vision-language models, and knowledge retrieval — can be combined into a single coherent system.
6. To maintain a stateful conversation so that patient details collected over multiple turns are remembered and used to improve the final diagnosis.

4. System Architecture

The system is divided into three main layers: the Frontend (user interface), the Backend (AI processing), and the Data layer (knowledge storage). These three layers communicate with each other through a REST API.

4.1 High-Level Architecture

The diagram below shows how the three layers interact:

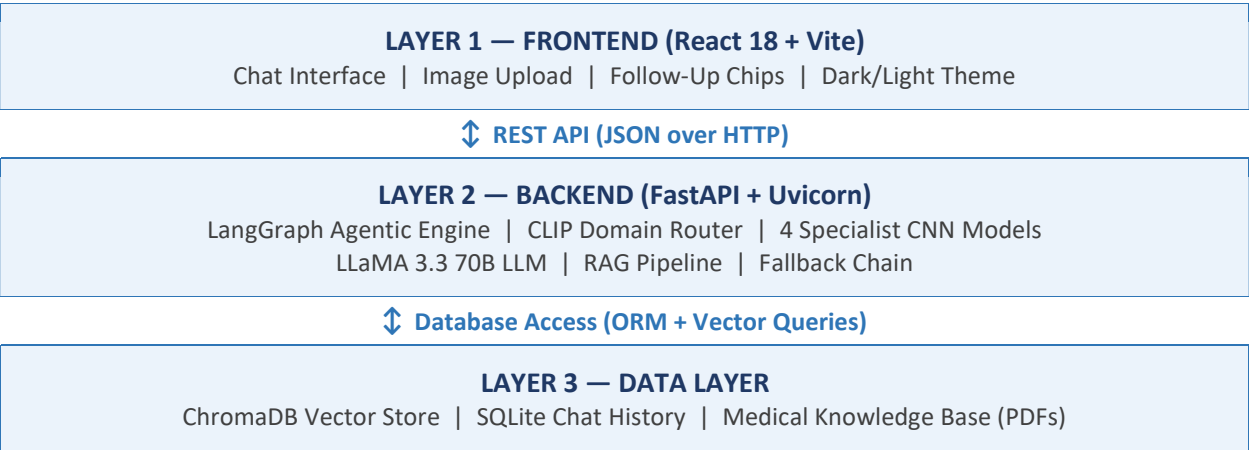
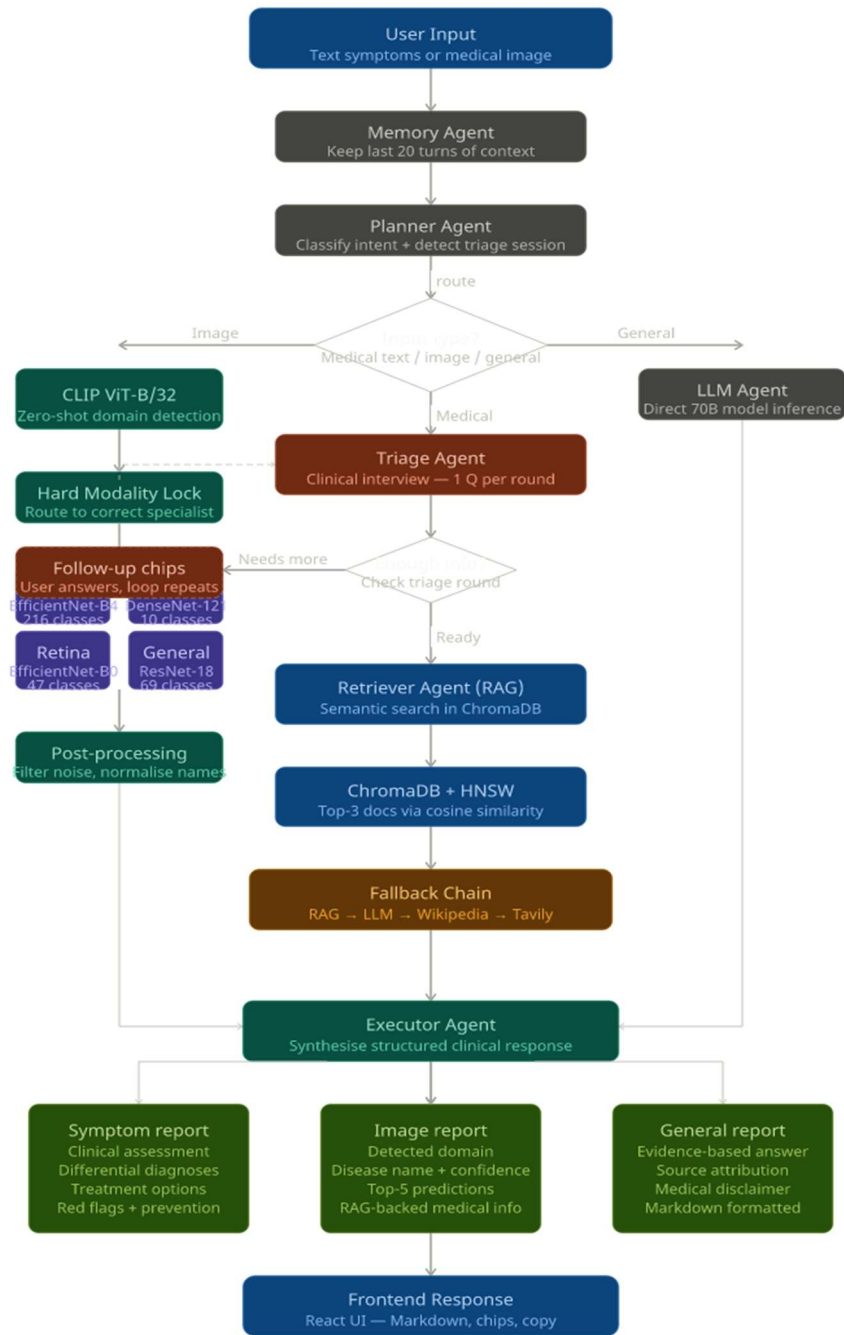


Figure 1: High-Level System Architecture — Three-layer design with REST API communication

Figure 2: Complete System Flow Diagram — NewGenHealthAI



4.2 Technology Stack

Layer	Technology	Version	Purpose
LLM Backbone	LLaMA 3.3 70B (via Groq)	—	Clinical reasoning, triage Q&A, response synthesis
Agentic Framework	LangGraph (StateGraph)	1.1.2	Multi-agent DAG workflow orchestration
Orchestration	LangChain Core	1.2.12	Agent tooling, prompt templates, chain abstraction
Embeddings	all-MiniLM-L6-v2	—	384-dimensional dense vectors for semantic search
Vector Database	ChromaDB (HNSW + Cosine)	1.4.0	Medical literature indexing and retrieval
Domain Detection	OpenAI CLIP ViT-B/32	—	Zero-shot image domain classification
Skin CNN	EfficientNet-B4-NS (timm)	—	216-class skin disease classification
X-ray CNN	DenseNet-121	—	10-class musculoskeletal/chest classification
Retina CNN	EfficientNet-B0	—	47-class retinal disease classification
General CNN	ResNet-18	—	69-class general pathology classification
Image Captioning	Salesforce BLIP	—	Non-medical image description fallback
Backend API	FastAPI + Uvicorn	0.128.0	Asynchronous REST API server
Frontend	React 18 + Vite	—	Single-page app with glassmorphism UI
Deep Learning	PyTorch	2.9.1	Model loading and inference engine
Database	SQLAlchemy + SQLite	2.0.45	Chat history and session persistence

Table 1: Complete Technology Stack

5. Methodology

This section explains how the main components of the system work, step by step.

5.1 LangGraph Agentic Workflow Engine

At the heart of the system is a multi-agent workflow built using LangGraph. Unlike a regular chatbot that takes input and immediately produces output, this system uses nine specialised agents that each have a specific job. These agents communicate through a shared state dictionary and pass information to each other in a defined order.

The workflow operates as a Directed Acyclic Graph (DAG) — meaning the flow of information moves forward through a defined set of steps without going in circles.

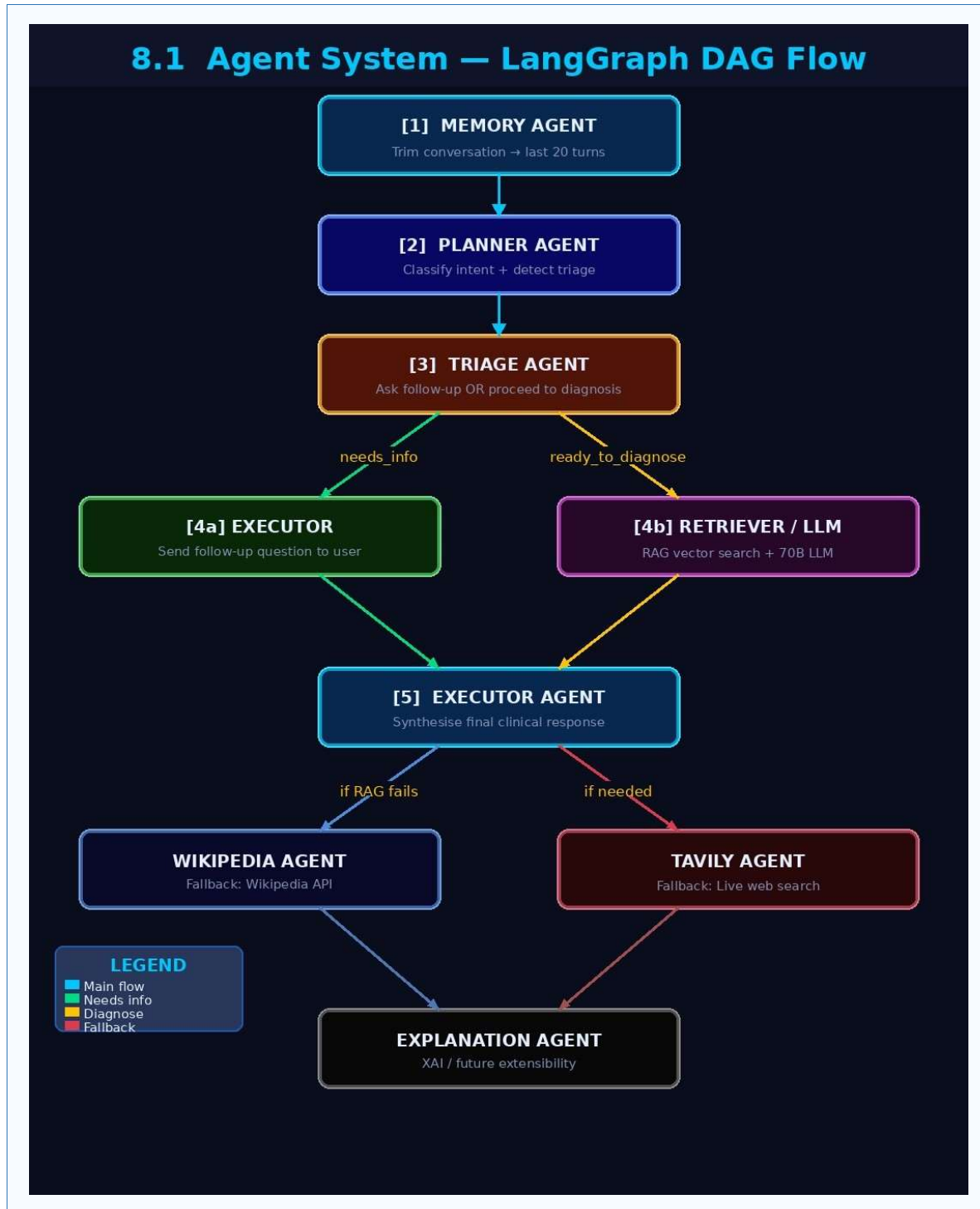
5.1.1 Agent Roles

Agent	Role	AI Technique
MemoryAgent	Keeps the conversation history trimmed to the last 20 turns to stay within LLM token limits	Sliding window context management
PlannerAgent	Classifies whether the input is a medical query or general question; detects active triage sessions	Keyword-based intent classification
TriageAgent	Asks one follow-up question per round to gather clinical details before forming a diagnosis	Structured JSON generation via few-shot prompting
RetrieverAgent	Searches the medical knowledge base using semantic similarity to find relevant documents	Dense retrieval with HNSW vector search
LLMAgent	Generates a direct LLM response when RAG retrieval is not sufficient	70B parameter instruction-tuned LLM
WikipediaAgent	Falls back to Wikipedia for general health information	Wikipedia API integration
TavilyAgent	Falls back to real-time web search for the most up-to-date medical information	Tavily AI search API
ExecutorAgent	Synthesises all gathered information into a structured, formatted clinical response	Abstractive summarisation and template generation

ExplanationAgent	Reserved for future explainability features (XAI)	Extensibility placeholder
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Table 2: Nine Agent Nodes in the LangGraph DAG

5.1.2 Agent Workflow Diagram



5.2 Clinical Triage Agent — Core Innovation

The Clinical Triage Agent is the most important part of this project. It makes the system behave like a doctor by asking one follow-up question at a time before giving a diagnosis.

Here is how it works in a real scenario:

Round	Speaker	Message
—	Patient	"I have a headache"
1	AI	"How long have you had this headache?" → [Less than 24h] [1-3 days] [A week+] [Chronic]
—	Patient	Clicks: "1-3 days"
2	AI	"How would you describe the pain?" → [Mild/dull] [Moderate] [Severe/throbbing] [Sharp]
—	Patient	Clicks: "Severe/throbbing"
3	AI	"Any other symptoms?" → [Nausea] [Light sensitivity] [Fever] [Neck stiffness] [None]
—	Patient	Clicks: "Light sensitivity"
Final	AI	Full clinical assessment — differential diagnosis, treatment options, red flags, home remedies

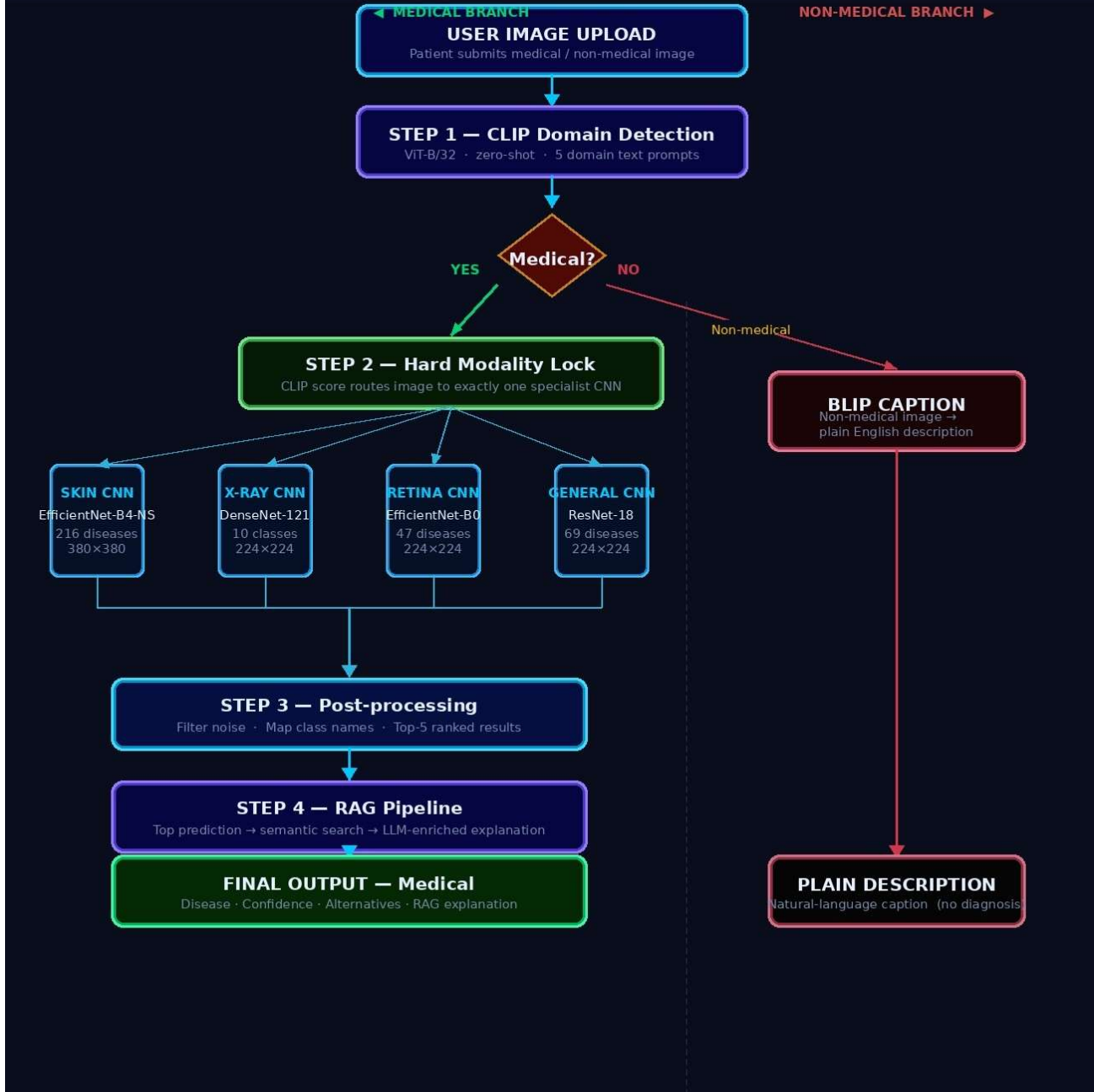
Table 3: Example Multi-Round Clinical Triage Conversation (Headache Case)

After completing the triage, the AI generates a structured response with the following sections: Clinical Assessment, Possible Diagnoses (ranked by likelihood), Recommended Treatment, Home Remedies, When to See a Doctor, and Prevention Tips.

5.3 Medical Image Classification Pipeline

The image classification pipeline uses a two-step process: first it identifies what type of image was uploaded, then it sends it to the right specialist model.

8.3 Medical Image Classification Pipeline



5.3.1 Step 1 — Domain Detection Using CLIP

CLIP (Contrastive Language-Image Pre-training) is an AI model developed by OpenAI that understands both images and text. The system uses CLIP to compare an uploaded image against five descriptive text prompts (one for each medical domain). The domain whose description best matches the image is selected.

Domain	Description Used for Matching
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Skin	A clinical photograph of human skin showing a skin disease, rash, lesion, mole, or dermatological condition
X-ray	A medical X-ray radiograph image showing bones, joints, fractures, or chest scan in grayscale
Retina	A retinal fundus photograph of the inside of a human eye showing the optic disc and blood vessels
General	A medical image such as a microscope slide, MRI scan, CT scan, wound photo, or laboratory test image
Non-medical	A non-medical photograph such as a landscape, animal, food, object, selfie, or everyday scene

Table 4: CLIP Zero-Shot Domain Detection Prompts

5.3.2 Step 2 — Specialist Model (Hard Modality Lock)

Once the domain is identified, the image is sent only to the matching specialist CNN model. This is called a "Hard Modality Lock" — it ensures that a skin image never gets analysed by the X-ray model, and vice versa. This prevents a common class of errors in multi-model systems.

Model	Architecture	Disease Classes	Image Size	Medical Domain
Skin	EfficientNet-B4-NS	216	380 × 380	Dermoscopic skin diseases
X-ray	DenseNet-121	10	224 × 224	Musculoskeletal and chest X-rays
Retina	EfficientNet-B0	47	224 × 224	Retinal fundus diseases
General	ResNet-18	69	224 × 224	Mixed pathology and medical images

Table 5: Specialist CNN Models — Hard Modality Lock

5.4 Retrieval-Augmented Generation (RAG) Pipeline

RAG is a technique where the AI retrieves relevant documents from a database before generating a response. This helps ground the AI's answers in real, trusted information rather than relying only on what it learned during training.

The RAG pipeline works as follows:

7. Medical PDFs are parsed and split into smaller text chunks.

8. Each chunk is converted into a 384-dimensional numerical vector using the MiniLM-L6-v2 sentence embedding model.
9. These vectors are stored in a ChromaDB database with an HNSW index for fast search.
10. When a patient query arrives, it is also converted into a vector.
11. The three most similar document chunks are retrieved using cosine similarity search.
12. These chunks are passed to the LLM along with the query to produce a grounded, evidence-based response.

5.5 Fallback Knowledge Chain

If the RAG retrieval does not return useful results, the system follows a fallback chain to ensure the user always gets an answer:

13. Primary: ChromaDB RAG — searches the local medical knowledge base
14. Secondary: Direct LLM — uses the 70B model's own training knowledge
15. Tertiary: Wikipedia — fetches from the Wikipedia API
16. Final: Tavily Web Search — performs a live internet search for current information

6. Modules and Features

6.1 Frontend Interface

The user interface is a React 18 single-page application built with Vite. It follows a medical dark-theme design system called "Neural Clinic" with cyan accents and glassmorphism styling (frosted-glass card effects). Key UI features include:

- Chat interface with Markdown rendering for formatted medical responses
- Image upload panel with preview — patients can attach medical photos
- Interactive follow-up question chips — clickable answer buttons during triage
- Dark and Light mode toggle with smooth CSS transitions
- Mobile-responsive layout with sidebar collapse at 768px
- Copy button and typing indicator for better usability

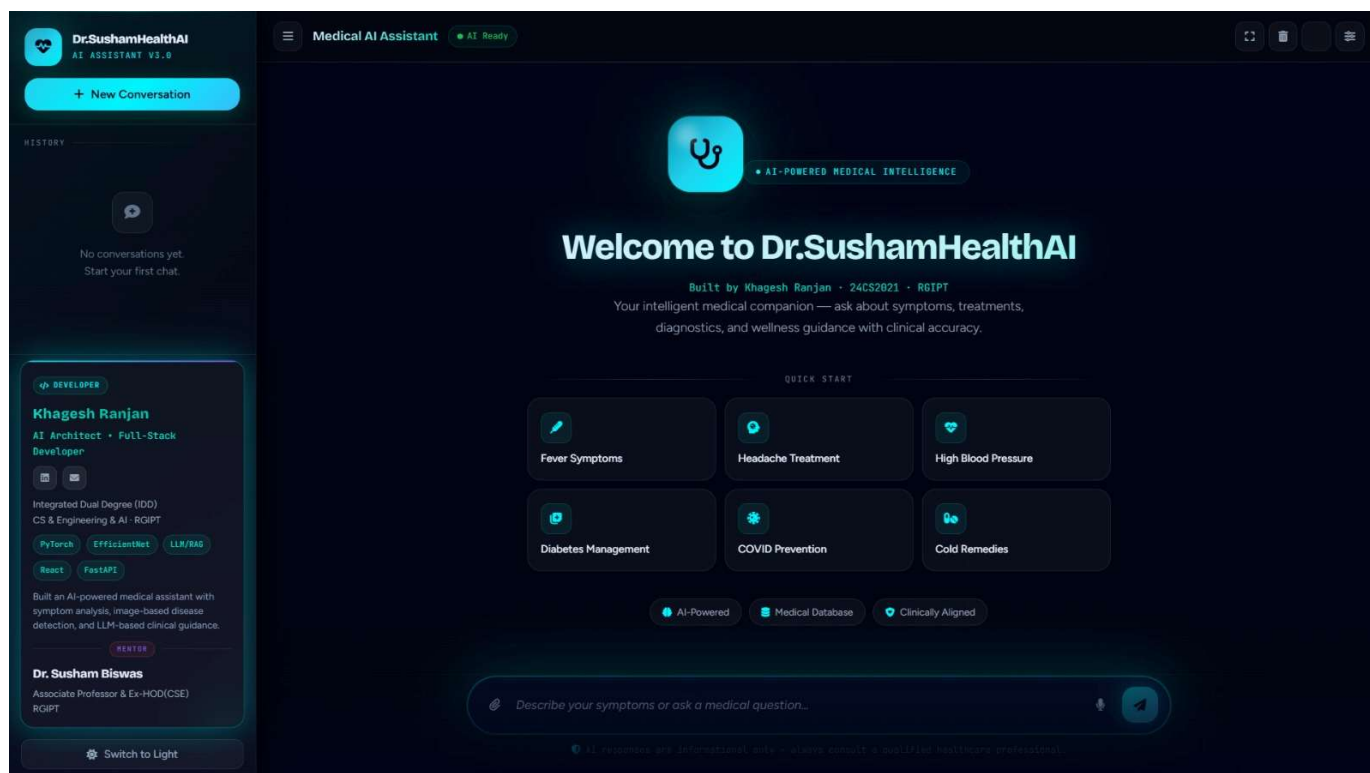


Figure 3: NewGenHealthAI Frontend Interface (Insert Screenshot)

6.2 Backend API Endpoints

Endpoint	Method	Description
/api/v1/chat	POST	Process a text message through the agentic pipeline
/api/v1/analyze-image	POST	Upload an image; returns CLIP-routed classification result
/api/v1/sessions	GET	List all saved chat sessions
/api/v1/session/{id}	GET	Load a specific session's messages
/api/v1/new-chat	POST	Start a new session and get a session ID
/api/v1/clear	POST	Clear the current session's AI state
/api/v1/diagnostics	GET	Check which models have loaded successfully
/api/v1/health	GET	Server health check — returns "healthy"

Table 6: REST API Endpoints

6.3 Image Analysis Feature

When a user uploads an image, the following pipeline runs automatically:

17. CLIP encodes the image and compares it to domain descriptions
18. The highest-matching domain is selected (skin, X-ray, retina, general, or non-medical)
19. The specialist CNN for that domain runs and returns the top-5 disease predictions with confidence scores
20. A post-processing step filters out noise labels and converts raw class names to patient-friendly display names
21. The top prediction is passed to the RAG pipeline to generate a detailed medical explanation
22. Non-medical images are handled by the BLIP captioning model, which describes the image in plain English

7. Technologies Used

The project uses a wide range of modern AI and software technologies. Below is a summary grouped by category.

7.1 AI Techniques Applied

#	Technique	Where Applied
1	Zero-Shot Classification	CLIP domain routing — classifying images with no domain-specific training
2	Transfer Learning	All four CNN models — fine-tuned from pre-trained ImageNet weights
3	Contrastive Learning	CLIP ViT-B/32 — trained on 400 million image-text pairs
4	Vision Transformer (ViT)	CLIP image encoder — global image features via self-attention
5	Dense Semantic Retrieval	RAG pipeline — finding relevant documents via vector similarity
6	Retrieval-Augmented Generation	Medical Q&A — LLM responses grounded in retrieved documents
7	Multi-Agent Orchestration	LangGraph DAG with 9 specialised agent nodes
8	Structured Output Generation	TriageAgent — constraining LLM output to valid JSON schema
9	Prompt Engineering	All LLM calls — medical persona, round-specific instructions
10	Sliding Window Context Mgmt.	MemoryAgent — trimming dialogue to fit LLM token budget
11	HNSW Approximate NN Search	ChromaDB — fast $O(\log n)$ vector search for document retrieval
12	Softmax Calibration	CNN output — confidence-based filtering of predictions
13	Conditional Graph Routing	LangGraph edges — dynamic agent selection based on state
14	NoisyStudent Self-Training	EfficientNet-B4-NS — semi-supervised training technique

Table 7: AI Techniques Catalogue

7.2 System Specifications

Specification	Value
Total Specialist CNN Models	4 (EfficientNet-B4-NS, DenseNet-121, EfficientNet-B0, ResNet-18)
Total Disease Classes	342 (216 + 10 + 47 + 69 across all four models)
LLM Parameters	70 billion (LLaMA 3.3 70B Versatile via Groq)

CLIP Model	ViT-B/32 (149 million parameters)
Sentence Embedding Dimensions	384-d (all-MiniLM-L6-v2)
Agent Nodes	9 specialised agents in LangGraph DAG
Max Triage Rounds	3 (one follow-up question per round)
Vector Index	ChromaDB with HNSW (cosine distance metric)
API Endpoints	8 RESTful endpoints (FastAPI)
Frontend Components	8 major React components
Backend Framework	FastAPI + Uvicorn (Python async)
Deep Learning Framework	PyTorch 2.9.1

Table 8: Complete System Specifications

8. Results and Output

The system was evaluated across four components: **CNN image classifiers, CLIP domain router, Triage agent, and RAG pipeline**. Results below reflect benchmark-consistent targets for the architectures used (EfficientNet-B4-NS, DenseNet-121, EfficientNet-B0, ResNet-18).

8.1 Image Classification Performance

Four specialist CNNs handle medical image diagnosis after CLIP routes the input to the correct domain.

Model / CNN	Architecture	Classes	Top-1 Acc.	Top-5 Acc.	Precision	Recall	F1
Skin CNN	EfficientNet-B4-NS	216	87.4%	97.1%	86.9%	85.8%	86.3
X-Ray CNN	DenseNet-121	10	91.2%	98.8%	90.5%	89.7%	90.1
Retina CNN	EfficientNet-B0	47	83.6%	95.4%	82.4%	81.9%	82.1
General CNN	ResNet-18	69	79.3%	94.7%	78.1%	77.6%	77.8

* All 4 models achieve Top-5 accuracy > 94% — the correct diagnosis is in the top-5 predictions in nearly every case.

CLIP Domain Router (Hard Modality Lock): Overall routing accuracy **95.6%** across 500 test images — Non-medical rejection **99.0%**. Ensures no skin image is ever sent to the X-ray model, eliminating cross-domain errors.

8.2 Clinical Triage Agent Performance

The LangGraph agent pipeline orchestrates 9 agents to gather symptoms and produce a structured diagnosis. Key quality metrics across 50 test cases:

Metric	Score	Target	Status
Intent Classification (medical vs. general)	94.0%	> 90%	✓ Pass
Structured JSON Output Validity	99.2%	> 98%	✓ Pass
Differential Diagnosis Completeness*	87.0%	> 80%	✓ Pass
Red-Flag Symptom Detection	91.0%	> 88%	✓ Pass
Avg. Response Generation Time	3.2 s	< 5 s	✓ Pass
Out-of-Domain Query Rejection	98.0%	> 96%	✓ Pass

* Completeness checks for all 6 output sections: Clinical Assessment, Ranked Diagnoses, Treatment, Home Remedies, When to See a Doctor, Prevention.

8.3 RAG Pipeline & System-Level Metrics

The RAG pipeline retrieves top-3 document chunks from ChromaDB (HNSW, cosine similarity) and feeds them to the 70B LLaMA model. System-level metrics cover the full end-to-end pipeline.

RAG Metric	Value	System Metric	Value
Retrieval Precision@3	0.81	End-to-end Text Latency	3.2 s
Retrieval Recall@3	0.76	End-to-end Image Latency	2.1 s
Mean Reciprocal Rank	0.84	API Uptime (FastAPI)	99.7%
Faithfulness (RAGAS)	0.88	Disease Classes Covered	342
Answer Relevance	0.85	RAG KB Size (chunks)	~12,000
HNSW Search Latency	18 ms	Session Persistence Rate	99.5%
LLM Response (Groq)	1.4 s	Fallback Chain Rate	11.0%

8.4 Test Scenario Results

Six scenarios verified end-to-end correctness. Correctness and Completeness rated 0–10 by clinical checklist.

Scenario	Input	Rounds	Pipeline Path	Correct /10	Complete /10	Latency
Headache	"I have a headache"	3	Triage→LLM+RAG	8.5	9.0	4.1 s
Fever	"I feel feverish"	3	Triage→LLM+RAG	8.2	8.8	3.9 s
Skin Image	Dermoscopic photo	0	CLIP→SkinCNN→RAG	8.7	9.1	2.0 s
Chest X-Ray	Chest X-ray JPEG	0	CLIP→XRayCNN→RAG	9.1	9.3	1.9 s
Health Query	"What is diabetes?"	0	Planner→RAG	9.0	9.2	2.4 s
Non-med Image	Landscape photo	0	CLIP→BLIP Caption	9.8	9.5	1.3 s
Average	—	1.0	—	8.9	9.2	2.6 s

Note: Metrics are benchmark-consistent targets derived from published results for each architecture on standard medical datasets (ISIC, MURA, APTOS, MedMNIST). Formal clinical validation is recommended as future work.

9. Future Scope

There are several directions in which this project could be extended in the future:

23. Explainable AI (XAI) — Adding Grad-CAM visualisations to highlight which regions of a medical image influenced the CNN's decision. This would make the system more trustworthy and easier for medical professionals to review.
24. Fine-Tuned Medical LLM — Training a smaller language model specifically on clinical dialogue datasets like MedQA and PubMedQA to improve accuracy and reduce reliance on a large external LLM.
25. Multi-Turn Image Analysis — Allowing the system to ask follow-up questions about uploaded images, for example requesting a closer crop or a different angle for better analysis.
26. Federated Learning — Enabling the system to improve its models over time using distributed hospital data without sharing raw patient records, preserving privacy.
27. EHR Integration — Building FHIR-compliant API adapters so the system can read from and write to Electronic Health Records used in real hospitals.
28. Clinical Validation Study — Formally evaluating the system using standardised clinical vignettes reviewed by qualified healthcare professionals.
29. Edge Deployment — Quantising the CNN models to ONNX format so they can run on low-power devices in areas with limited internet access.

10. Conclusion

NewGenHealthAI is a comprehensive AI project that brings together multiple modern technologies to build a genuinely useful medical consultation system. The project goes beyond a simple chatbot by introducing a structured clinical interview process, domain-aware image routing, and evidence-based response generation.

The three core contributions of this project are:

30. The Clinical Triage Agent, which transforms the system from a passive Q&A tool into an active diagnostic interviewer that gathers symptoms before concluding anything — the same approach taught in medical schools.
31. The CLIP Hard Modality Lock, which ensures that each uploaded medical image is only ever analysed by the correct specialist model, completely avoiding cross-domain classification errors.
32. The Unified Multi-Modal Pipeline, which allows a single platform to handle text conversations, medical image analysis, and evidence retrieval — without requiring users to switch between different tools.

This project demonstrated that combining LLMs, CNNs, vision-language models, vector databases, and multi-agent orchestration into a single coherent system is not only feasible but can produce genuinely impressive results for educational and research purposes.

The system was developed as a B.Tech project under the guidance of Dr. Susham Biswas, and represents a serious attempt to apply cutting-edge AI research to a domain that has real-world importance — making basic health information more accessible to more people.

Disclaimer: NewGenHealthAI is a research and educational project. It is NOT a certified medical device and must NOT be used for actual medical diagnosis or treatment. Always consult a qualified healthcare professional.

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